

Week 2 – Deep Learning with PyTorch

AI/ML Internship | Deep Learning Model-Building Phase

1. Objective

The objective of Week 2 was to develop practical deep-learning skills using PyTorch. The work covered feed-forward neural networks, convolutional neural networks (CNNs), model training and evaluation, controlled hyperparameter experimentation, and comparison of model performance.

2. Dataset and Tools

Dataset: MNIST handwritten digit image dataset. The dataset contains grayscale 28×28 pixel images belonging to ten digit classes (0–9).

Tools: Python, PyTorch, Torchvision, NumPy, Pandas, Matplotlib, Scikit-learn, and Google Colab.

3. Task 2.1 – Feed-Forward Neural Network

A feed-forward neural network was implemented in PyTorch. The image data was prepared using datasets and DataLoaders, and the 28×28 images were flattened into input features. The network used fully connected layers with ReLU activation and produced ten output classes. Cross-entropy loss was used for classification and Adam was used for optimization. Training and validation performance were monitored and the trained model was evaluated on the test dataset.

4. Task 2.2 – CNN Image Classifier

A convolutional neural network was developed for MNIST image classification. The CNN used convolutional layers, ReLU activations, max-pooling, a fully connected layer, and dropout before the final classification layer. The model was trained with Adam and evaluated using test accuracy, a classification report, a confusion matrix, and sample predictions. These evaluation outputs were included in the Week 2 Colab notebook.

5. Task 2.3 – Hyperparameter Experimentation

Controlled experiments were performed to observe how training settings affected model performance. The experiments varied the optimizer, learning rate, and batch size while keeping the comparison structured. The notebook records the configurations and resulting performance in an experiment comparison table and visualization.

Experiment	Optimizer	Learning Rate	Batch Size	Epochs
1	Adam	0.001	64	2
2	Adam	0.01	64	2
3	SGD	0.001	64	2
4	Adam	0.001	128	2

6. Findings and Learning Outcomes

The experiments demonstrated that training configuration can influence model performance and training behavior. The Week 2 work strengthened practical understanding of PyTorch tensors, datasets, DataLoaders, loss functions, optimization, neural-network architecture, CNN-based image classification, evaluation metrics, and controlled experimentation. The comparison results are available directly in the Colab notebook.

7. Limitations and Future Improvements

The experiments were designed for a practical internship workflow and used a limited number of training configurations and epochs. Future work could include longer training, broader hyperparameter search, learning-rate scheduling, data augmentation, additional CNN architectures, and evaluation on more challenging image datasets.

8. Conclusion

Week 2 successfully covered the required deep-learning tasks using PyTorch. A feed-forward network and CNN classifier were implemented and evaluated, controlled hyperparameter experiments were conducted, and the findings were documented. The complete implementation, outputs, experiment comparisons, and supporting results are contained in the Week 2 Colab notebook.

Primary implementation file: AbdUrRahman_Week2_Deep_Learning_PyTorch.ipynb